

AI-POWERED GEOLOGICAL, MINING & REPORTING ECOSYSTEM

Detailed Project Description | CMPDI and Coal India Subsidiaries

A unified platform for intelligent document extraction, geological data structuring, automated topic discovery, evidence-grounded query answering and human-verified administrative reporting.

Prepared from: the supplied 15-page “Strategic Blueprint for an AI-Powered Geological, Mining, and Reporting Ecosystem at CMPDI and CIL Subsidiaries.”

Document scope: source-derived project design, with clearly marked implementation recommendations and cautions. The project is a proposal, not a claim that the system has already been deployed.

1. Executive overview

The Central Mine Planning and Design Institute (CMPDI) generates and works with geological exploration, mine-planning and environmental records. Coal India Limited (CIL) subsidiaries and administrative teams also handle production data, grievances, committee requests and parliamentary questions. Much of the information needed for a single response is distributed across scanned reports, borehole logs, spreadsheets, archives and enterprise systems.

The proposed solution consolidates these materials into an AI-assisted reporting ecosystem. Its three principal capabilities are (1) multimodal document ingestion and structured extraction, (2) topic identification and word-cloud analytics, and (3) retrieval-augmented generation (RAG) for evidence-based questions and draft reports. A spatial PostgreSQL/PostGIS data layer, human verification, enterprise integrations and secure model serving support these capabilities.

2. Problem statement and operational need

Manual preparation of technical and administrative responses requires staff to locate documents, interpret specialized tables, reconcile figures and draft answers. Conventional character recognition may lose the relationship between a borehole depth interval, its lithological description and a measurement in the same row. Separate ERP databases are suitable for transactions but do not themselves interpret unstructured geological figures or handwritten logs.

- Heterogeneous source formats: searchable PDFs, image-only scans, handwritten or typed logs, spreadsheets, geological charts and narrative reports.
- Fragmented data across exploration teams, subsidiaries, administrative archives and CoalNet/SAP exports.
- Difficult extraction of coal seam thickness, lithology, ash content, gross calorific value (GCV), moisture and coordinates from irregular layouts.
- High verification burden for time-sensitive parliamentary, ministerial and committee reporting.
- Multilingual and code-mixed Hindi/English material that can challenge generic text pipelines.
- Limited thematic visibility into large volumes of complaints, operational reports and correspondence.

3. Objectives and intended users

Objective	Expected capability
Digitize archives	Convert heterogeneous pages and tables into validated, structured records with source traceability.
Geological intelligence	Relate boreholes, strata and coordinates for retrieval and downstream geospatial analysis.
Topic analytics	Group documents by emerging themes and display representative keywords and word clouds.
Answer generation	Produce draft answers grounded in retrieved CMPDI/CIL records with citations.
Human verification	Require domain-expert review for sensitive or uncertain results before dissemination.
Scalable operations	Serve subsidiary users with controlled access, logging and enterprise integration.

Intended users: geologists, mine-planning teams, subsidiary nodal officers, administrative analysts, Parliamentary Affairs personnel, compliance teams and authorized leadership. Access privileges should vary by role.

4. Functional modules

4.1 Intelligent document ingestion and processing

The ingestion layer accepts document batches and assigns immutable document identifiers, provenance, timestamps, version information and access classifications. For digitally generated PDFs, embedded text can be extracted directly; image-only and visually complex pages are routed to layout analysis and vision-language processing. This routing distinction is an implementation recommendation to reduce cost and avoid unnecessary OCR.

The source proposes layout-aware models such as LayoutLMv3 for text-image-layout relationships, ColPali for visual document retrieval, SmolDocling for structured key-value extraction and RT-DocLayout for region detection. These are candidate approaches, not a requirement to run all models simultaneously. Model selection should follow a benchmark on representative CMPDI pages.

Processing should preserve page number, bounding region and original document reference for each extracted fact. Geological rows must retain their depth interval and associated column headings; extracted values should not be detached from the source context.

4.2 Geological table and borehole-log extraction

Borehole logs describe the vertical subsurface profile using depth intervals, lithological observations, sample identifiers and measurement columns. The pipeline must reconstruct table boundaries, distinguish header rows from data rows, normalize measurement units and validate interval continuity. Charts or diagrams that cannot be reliably translated into a structured record should remain accessible as linked page images.

Extracted field	Validation / representation
Borehole identifier	Resolve to a unique borehole and associated report.
Location and coordinates	Store coordinate reference system and check plausible ranges.
From depth / to depth	Numeric interval; ensure end depth is greater than start depth.
Lithology / seam	Controlled vocabulary with retained original text.
Seam thickness	Derived from validated intervals where appropriate.
Ash, GCV, moisture	Preserve units, sampling method and source page.
Source evidence	Document ID, page, region and reviewer status.

4.3 Indic-language and audio support

The source discusses India-oriented multimodal AI, including Sarvam Vision for document understanding, Saaras V3 for speech recognition and Bulbul V3 for text-to-speech. These are optional integration candidates for Hindi, English and regional-language material. Transcription quality and availability must be tested against actual subsidiary documents and recordings before deployment.

4.4 Spatial database and geological retrieval

Validated records are normalized into PostgreSQL with the PostGIS extension. The source proposes GiST spatial indexes to accelerate location-based retrieval. A user should be able to search for boreholes in a specified region, inspect their logs and retrieve related geological or environmental reports. Spatial analysis such as 3D correlation or Kriging is a downstream use case; it requires appropriate geological controls and is not automatically produced by document extraction alone.

5. Topic identification and word-cloud analytics

The second major module applies BERTopic to large document collections such as incident reports, inter-ministerial correspondence, operational summaries and parliamentary questions. The source identifies the pipeline as transformer embeddings, UMAP dimensionality reduction, HDBSCAN clustering and class-based TF-IDF (c-TF-IDF) topic representation.

Pipeline stage	Purpose
Document embeddings	Represent semantic similarity beyond exact keyword matches.
UMAP	Reduce embedding dimensionality for subsequent clustering.
HDBSCAN	Group semantically related records and identify noise/outliers.
c-TF-IDF	Extract distinctive words for each discovered topic.
Topic dashboard	Display topic labels, document counts, trends and word clouds.

The module can help analysts detect recurring subjects in Khanan Prahari complaints or administrative correspondence. Topic clusters and word clouds are exploratory summaries, not verified conclusions about incidents or compliance. The source recommends topic coherence measures C_V and normalized pointwise mutual information (C_NPMI); interpretation should also be checked by domain reviewers.

6. Retrieval-augmented question answering

The third core module provides an evidence-grounded interface for questions such as: “What does the latest approved report state about a specified coal seam?”, “Which source supports a production figure?” or “Prepare a draft response to a parliamentary question using approved records.” The RAG workflow retrieves relevant source material before generating a response.

- Query intake: authenticate the user, determine scope and identify the requested reporting period, subsidiary and subject.
- Retrieval: search textual embeddings, structured records and, where relevant, visually indexed pages.
- Evidence selection: rank results and retain source identifiers, document dates and page references.
- Draft synthesis: instruct the language model to answer only from authorized, retrieved evidence.
- Numeric verification: compare critical production or financial figures with structured records and document dates.
- Review: display citations, uncertainty and unsupported assertions; send sensitive drafts to an authorized officer.
- Export: produce a reviewed response or report with provenance and an audit trail.

The source mentions a large Sarvam model as a possible generator. The deployment should select a model after testing factual grounding, Indic-language quality, latency, hosting constraints and cost. A long context window alone does not guarantee accuracy.

6.1 RAG evaluation and quality gates

Measure	What it checks
Context precision	Whether retrieved passages are relevant to the question.
Context recall	Whether necessary supporting evidence was retrieved.
Faithfulness	Whether generated claims are supported by retrieved material.
Answer relevance	Whether the response addresses the user question.
MRR / hit-rate	Whether relevant sources appear near the top of results.

The source proposes RAGAS for automated evaluation and emphasizes human-in-the-loop (HITL) approval. Automated scores should be treated as diagnostic indicators, not proof of zero hallucinations. Any official parliamentary or ministerial answer requires the responsible human authority to verify facts and approve release.

7. End-to-end technical architecture

Layer 1 — Source systems: geological reports, borehole logs, environmental documents, scanned records, correspondence, Khanan Prahari records, and authorized CoalNet/SAP exports.

Layer 2 — Ingestion: connectors, batch uploads, file validation, document classification, access metadata and deduplication.

Layer 3 — Multimodal parsing: direct PDF extraction where possible; layout detection, VLMs and language-specific processing for complex pages; field-level validation and review queues.

Layer 4 — Storage: original files in controlled object storage; normalized facts in PostgreSQL/PostGIS; text and image retrieval indexes with provenance links.

Layer 5 — Intelligence: BERTopic analysis, hybrid retrieval, LLM-based draft generation and evaluation checks.

Layer 6 — Applications: analyst search, geological record viewer, topic dashboard, parliamentary answer drafting, document verification and exports.

Layer 7 — Governance: role-based authorization, encryption, approval workflow, model monitoring, audit logging and retention controls.

8. Data schema and example records

The source specifies PostgreSQL/PostGIS and JSON-to-relational mapping. The following schema is an illustrative implementation extension rather than an existing CMPDI database definition.

Entity	Illustrative fields
Document	document_id, title, category, owner, version, confidentiality, date
Page / region	document_id, page_no, bounding_region, text, extraction_method
Borehole	borehole_id, report_id, latitude, longitude, geometry
Lithology interval	borehole_id, from_depth, to_depth, lithology, seam_name

Entity	Illustrative fields
Coal analysis	sample_id, ash_pct, moisture_pct, gcv, test_unit
Topic assignment	document_id, topic_id, topic_score, model_version
Evidence chunk	chunk_id, document_id, page_no, text, embedding_ref
Response draft	question_id, draft_text, evidence_refs, reviewer, approval_status

Every extracted numerical value should retain its source and units. Conflicting measurements across document versions should be surfaced to reviewers rather than silently reconciled by the language model.

9. Human-in-the-loop and administrative workflow

Human review is essential at two checkpoints. During ingestion, geologists or trained operators verify low-confidence depth rows, tabular associations and ambiguous scans. During report generation, the relevant administrative or technical authority verifies evidence, dates, calculations and wording before an answer becomes official.

- Show the original page beside each extracted value and allow correction with a reason code.
- Maintain reviewer identity, timestamp and the before/after values for every correction.
- Route low-confidence or conflicting extractions into a queue rather than silently accepting them.
- Require approval for sensitive parliamentary, ministerial or external-facing outputs.
- Record which source versions supported each approved answer and invalidate drafts when critical sources change.

10. Enterprise infrastructure and security

The source recommends deployment on a MeitY-empanelled sovereign cloud provider and proposes NVIDIA Triton Inference Server for heterogeneous AI serving, with TensorRT-LLM optimizations for supported language-model workloads. These are architecture proposals and should be validated against procurement, security and performance requirements.

A production environment should separate document storage, database services, vector retrieval, inference services and the web application. Network segmentation, least-privilege access, encryption in transit and at rest, audit logs, backups and disaster recovery should be designed before live sensitive records are loaded. Integration with CoalNet and SAP should use authorized interfaces and read-only data access wherever feasible.

Performance engineering should benchmark concurrency, throughput, page-processing time and GPU memory. Triton may improve throughput for supported workloads, but neither zero latency nor universal compatibility can be assumed.

11. Implementation roadmap

Phase	Deliverables / acceptance focus
1. Requirements and sandbox	Inventory document types and access rights; choose sample corpus; define extraction schemas, security requirements and baseline tests.
2. Digitization and validation	Build PDF/VLM ingestion, provenance mapping, borehole schema, PostGIS persistence and reviewer correction interface.
3. Analytics and RAG	Add BERTopic, search, evidence-grounded drafting, citation viewer and RAGAS-based evaluation.
4. Enterprise integration	Connect authorized CoalNet/SAP data, configure secure inference and train geologists, nodal officers and reporting teams.
5. Pilot and iteration (recommended)	Pilot with a controlled dataset; benchmark quality and review effort; tune thresholds before expanding to more subsidiaries.

12. Validation strategy and acceptance criteria

The source proposes layout-aware intersection over union (IoU), topic coherence and RAGAS retrieval/generation measures. A realistic pilot should additionally measure field-level exact match, numerical tolerance, row association, end-to-end document processing time, reviewer correction rate, source-citation accuracy and official-draft turnaround time.

Build separate evaluation sets for born-digital PDFs, clean scans, degraded scans, complex borehole tables and Hindi/English documents. Report results by document type rather than presenting one aggregate accuracy figure. Test adversarial questions that request information not present in the approved corpus: the system should decline to fabricate a number and identify missing evidence.

13. Expected outcomes and project targets

The source document lists the following **projected targets**. They are not independently verified performance results or guaranteed outcomes; a pilot must establish actual baselines and measured improvements.

Metric	Source-projected target	Validation requirement
Report preparation time reduction	75–85%	Compare elapsed drafting and approval time on matched tasks.
Structured extraction accuracy	95–98%	Measure field-level correctness on labeled geological documents.
Repetitive workflow automation	80–90%	Define automatable tasks and quantify human intervention.

Additional intended benefits include searchable historical knowledge, improved consistency of cited figures, faster thematic triage and less manual transcription. Realized benefits will depend on source quality, access permissions, model selection and the quality of human review.

14. Risks, constraints and mitigations

Risk	Recommended mitigation
Poor scans and irregular tables	Use page-type routing, image retention, field-level confidence and specialist review.
Hallucinated or outdated answers	Restrict generation to approved evidence, show dates and citations, validate numbers and require approval.
Confidential information leakage	Enforce document-level authorization at retrieval time and isolate sensitive workloads.
Incorrect geological associations	Validate borehole identifiers, depth ranges, coordinates and source-linked table cells.
Multilingual extraction errors	Benchmark Indic-language documents and escalate uncertain translations.
Model or data drift	Monitor quality by document type, track model versions and retest after changes.
Integration complexity	Pilot with authorized exports before committing to live ERP connectors.

15. Demonstration scenario

Scenario: An authorized analyst receives a question requesting the latest approved coal-quality figures for a particular area. The analyst enters the area, period and requested metrics. The system searches structured borehole records and related geological reports, retrieves relevant tables, displays the source pages and drafts a response containing only verified values. A geologist checks seam and sample associations; the reporting officer verifies the final wording and authorizes export.

In parallel, an administrative analyst can inspect a BERTopic dashboard to identify frequent themes in a set of incident reports and open the original documents behind any cluster. These are proposed user journeys, not descriptions of a completed implementation.

16. Deliverables

- Secure document-ingestion interface with provenance and processing status.
- Validated geological extraction pipeline and normalized PostGIS data model.
- Original-document viewer with field-level verification and correction history.
- Searchable knowledge repository and source-linked RAG interface.
- Topic exploration dashboard with word clouds and document drill-down.
- Draft reporting workspace with citations, reviewer approval and export.
- Evaluation dashboard for extraction, retrieval, groundedness and latency.
- Deployment documentation, access controls, operations runbook and user training material.

17. Conclusion

The proposed CMPDI/CIL platform connects document intelligence, geological data management, thematic analytics and evidence-grounded administrative reporting. Its value lies not simply in generating text faster, but in preserving source provenance, structuring specialized geological records and making every official answer reviewable. The source blueprint proposes VLM-assisted extraction, PostgreSQL/PostGIS, BERTopic, RAG/RAGAS, HITL and secure high-throughput inference as the main building blocks. A phased pilot with domain-expert validation is the practical route from architecture to an operational system.

Source and interpretation notes

Primary source: user-supplied 15-page “Strategic Blueprint for an AI-Powered Geological, Mining, and Reporting Ecosystem at CMPDI and CIL Subsidiaries,” especially sections 1–8. Its appended works-cited list is not independently verified here. Figures, model names and projected percentages are attributed to that source; schema examples, quality controls, deployment cautions and the additional pilot phase are explicitly proposed implementation details.